

Nicola Saltarelli

 LinkedIn |  GitHub |  Bari, Italy

EDUCATION

Politecnico di Bari

Master degree in Automation Engineering

Oct. 2023 – Oct. 2025

Bari, Italy

- **GPA:** 3.92/4.0
- **Final grade:** 110/110 with honors

Politecnico di Bari

Bachelor degree in Computer and Automation Engineering

Oct. 2020 – Jul. 2023

Bari, Italy

- **Final grade:** 110/110 with honors

EXPERIENCE

Politecnico di Bari

Research Fellow

Nov. 2025 – Present

Bari, Italy

- Conducting R&D activities focused on producing scientific papers in the fields of Robotics, Artificial Intelligence, and Computer Vision.

Politecnico di Bari

MSc Research Intern (Master Thesis)

Jun. 2025 – Oct. 2025

Bari, Italy

- **Thesis:** "Online Reinforcement Learning for Adaptive Gain Tuning in Image-Based Visual Servoing"  GitHub
- Designed a ROS2 (Jazzy) / Gazebo environment for a UR5e and developed a Continuous Action Q-Learning (CAQL) algorithm for adaptive IBVS, integrating YOLO, AprilTag, and MoveIt Servo.

PROJECTS

LPBF thermal monitoring deep learning | Python, MATLAB YouTube

- Engineered a Computer Vision pipeline based on a modified ResNet-18 architecture to monitor Laser Powder Bed Fusion (LPBF) processes, classifying laser power and scanning speed from thermal images.

Turtlebot3 Autonomous Exploration | ROS 2, Gazebo GitHub

- Developed and simulated an autonomous exploration system for a TurtleBot3 robot leveraging ROS 2 and Gazebo.

Motion Control with Neural Network | Simulink, MATLAB GitHub

- Implemented a two-layer neural network for the motion control of a 6-DOF robotic manipulator (Puma 560), utilizing Simulink and Peter Corkes Robotics Toolbox.

Snake on FPGA DE10-Lite | VHDL GitHub

- Designed and deployed the classic Snake game on an FPGA DE10-Lite hardware platform utilizing VHDL.

Position Control of Linear Actuator | LabVIEW, C++ GitHub

- Engineered a closed-loop position control system for a linear actuator using an Arduino microcontroller and LabVIEW, designing and 3D-printing all custom components for the physical setup.

SKILLS

Coding: Java, Python, MATLAB, R, SQL, JavaScript, C++, HTML, CSS, NodeJS, ReactJS, VHDL

Tools: Simulink, Solidworks, LabView, Fritzing, NI Multisim, ROS2, Docker

Languages: Italian (Native), English (B2)